



CHEMISTRY Department
SIMHADRI

Note No :1



Ref:SMTTP/Chemistry/2025-26/Chem/226235

Date:13/06/2025(16:24:46)

Subject:LMI on Code of Practices on Power Plant Chemistry

OGN of Code of Practices on Power plant Chemistry was revised by CC-OS in April-2025. LMI of Simhadri LMI/OGN/OPS/CHEM/015 is revised in line with the new OGN.

Previous LMI Version	Present LMI Version
Issue 04, Rev:00, Jan-2023	Issue 04, Rev:01, June-2025

Changes in LMI are briefed in the below table. LMI & OGN are attached.

Sl.No.	Code of Practices	Modification
1.	Sec 4.1-General Guidelines	New points incorporated.
2.	Sec 4.2 table-1- IOM-Summary	New IOMs appended.
3.	Sec 5.1-RW/Clarified Water/Filter water	New parameters incorporated.
4.	Sec 5.2 on Resin-based DM Plant	Newly incorporated.
5.	Sec 5.2- Tables 3A, 3C	Newly incorporated.
6.	Sec 5.2- Common Guidelines for Resin-	Newly incorporated

	Based DM Plant	
7.	Sec 5.7 table 6B- Specific Chemical Consumption Data for DM Plant	Modified for calculation of existing ionic load-based specific chemical consumption
8.	Sec 6.0, table 8C-Main Plant parameters to be tested	Foot notes modified.
9.	Sec 8.0, table 9A & 9B-Coal Analysis	Foot notes modified.
10.	Sec 9.0- Limestone & Gypsum Analysis for FGD	Newly incorporated.
11.	Sec 10.0-B, table 10B & 10C- Lube oil Analysis	Mill hydraulic oil monitoring added. CHP oil frequency modified. One oil added in Gear Box oil section. Foot notes modified.
12.	Sec 10.0-C-DGA	Foot notes modified.
13.	Sec 13.0, table 13A-Online Instruments	Modified.
14.	Sec 14.0, table 14A- Equipment Inspection	Modified for checks to be done and frequency, foot notes modified.
15.	Sec 14.0, table 14C-Overhauling Jobs-Chemistry section	Description and responsibilities modified.
16.	Annexure-II- Clarified Water and DM Water Cost Calculation Template	Newly incorporated.

Put up for Kind approval of CA for implementation.

Submitted by:

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On:13/06/2025(16:24:46) **Ref:**SMTTPP/Chemistry/2025-26/Chem/226235

Note No:2

OGN on Code of practices on power plant Chemistry was revised by CC-OS. Hence LMI on Code of practices on power plant Chemistry updated with modified incorporations and submitted for approval of competent authority for implementation at NTPC Simhadri.

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Note No:3

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Note No:4

Please review.

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Note No:5

Following corrections are required

- 1.Checklist should be kept in last part of LM as Annexures
- 2.Last 3 pages Formatting as per standard format to be done.
3. LMI No. should not contain issue 04,It is to be corrected at each page.

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Note No:6

Corrections are done and draft LMI attached.

Put up for Kind approval of CA for implementation.

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Note No:7

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Note No:8

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Note No:10

LMI Reviewed

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Note No:11

LMI is in line with the OGN.

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Note No:12

pl review MM offsite functions.

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Note No:13

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Note No:14

LMI is reviewed by MM OFS and forwarded please.

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Note No:15

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Note No:17

Safety and FQA may please review the attached LMI.

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Note No:20

LMI is in line with the OGN. May please be approved.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Code of Practices on Power Plant Chemistry

LMI/OGN/OPS/CHEM/015
Revision: 01, Date: June-2025

Approved for Implementation by: HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Code of Practice on Power Plant Chemistry
2	LMI No.	LMI/OGN/OPS/CHEM/015 revision 01
3	Reference Document	COS-ISO-00-OGN/OPS/CHEM/015 Rev. No.: 04 Apr. 2025
4	Superseded Document	LMI/OPS/CHEM/015 Issue 04 and revision 00
5	Prepared By	Name: Devesh Kumar Manager (CHEM)
6	Checked By	Name: Santanu Jana DGM (CHEM)
7	Reviewed By	Name: P K Jena GM (O&M)
8	Approved By	Name: Sameer Sharma H O P (Simhadri)

TITLE	Code of Practices on Power Plant Chemistry
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1.0 Introduction

The document "Code of Practice on Power Plant Chemistry" covers chemistry parameters to be tested/ recorded and their frequencies for water treatment plant, steam water cycle, coal, oil, gas, cooling water system. Besides this, the document contains list of various laboratory instruments for testing water, coal, oil, gas etc., online instruments for WTP and steam water cycle monitoring, reporting templates and equipment inspection details. The various parameters and testing/recording frequencies pertaining to WTP, steam water cycle, coal, oil, gas, resins etc. have been optimized keeping in view the requirements of specific fleet of units. Summary of guidelines issued by COS has been appended in the current version. Overhauling checklist has been newly incorporated.

This document shall be of immense use for day-to-day operational control and as a reference in the area of power plant chemistry.

2.0 Superseded Documents

Code of practice on Power Plant Chemistry LMI/OGN/OPS/CHEM/015 Issue 04 and revision 00

3.0 Details of revision

Sl.No.	Code of Practices	Modification
1.	Sec 4.1-General Guidelines	New points incorporated.
2.	Sec 4.2 table-1- IOM-Summary	New IOMs appended.
3.	Sec 5.1-RW/Clarified Water/Filter water	New parameters incorporated.
4.	Sec 5.2 on Resin-based DM Plant	Newly incorporated.
5.	Sec 5.2- Tables 3A, 3C	Newly incorporated.
6.	Sec 5.2- Common Guidelines for Resin-Based DM Plant	Newly incorporated
7.	Sec 5.7 table 6B- Specific Chemical Consumption Data for DM Plant	Modified for calculation of existing ionic load-based specific chemical consumption
8.	Sec 6.0, table 8C-Main Plant parameters to be tested	Foot notes modified.
9.	Sec 8.0, table 9A & 9B-Coal Analysis	Foot notes modified.
10.	Sec 9.0- Limestone & Gypsum Analysis for FGD	Newly incorporated.
11.	Sec 10.0-B, table 10B & 10C- Lube oil Analysis	Mill hydraulic oil monitoring added. CHP oil frequency modified. One oil added in Gear Box oil section. Foot notes modified.
12.	Sec 10.0-C-DGA	Foot notes modified.

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13.	Sec 13.0, table 13A-Online Instruments	Modified.
14.	Sec 14.0, table 14A- Equipment Inspection	Modified for checks to be done and frequency, foot notes modified.
15.	Sec 14.0, table 14C-Overhauling Jobs- Chemistry section	Description and responsibilities modified.
16.	Annexure-II- Clarified Water and DM Water Cost Calculation Template	Newly incorporated.

4.0 Guidelines

4.1 General Guidelines

- Sites shall maintain design parameters/ data (as annexures) for the following, in their station- level LMI (as applicable), corresponding to this OGN:
 - Design and current raw water analysis parameters as per parameters template given in 5.0.
 - Drinking water specifications as per IS 10500 (latest version)
 - Design coal analysis parameters- For as-fired coal; Mines-wise declared grade for as- received indigenous coal; specifications of as-received imported coal as per tender documents.
 - Specifications for fuel oil viz. LDO, HFO, HSD
 - Turbine lube oil specifications (as per IS and/or OEM-specified)
 - Control fluid specifications (OEM specified).
 - Fans, APH, Mills- lube oil specifications (OEM-specified)
 - Specifications of CHP oils (OEM-specified)
- Testing frequency of all parameters are to be maintained as defined in this TC document; however, frequency may be **increased** based on site-specific requirement.
If lab analysis is adversely affected due to instrument break-down or non-availability, the samples are to be tested from CC-NETRA or external Govt/CSIR/NABL-accredited labs or other NTPC labs.
- Station may procure additional instruments (other than those mentioned in this TC document) based on site-specific requirements. However, all instruments as per list are to be available with site or site has to have separate provisions or service contract for external testing as per frequency mentioned in this OGN.
- All critical issues of the Chemistry section are to be highlighted through ORT exception points for review.
- Chemistry Benchmarking Score is to be evaluated and the document is to be

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uploaded on monthly basis (stagewise) on Chemistry digitization dashboard of “Advanced Analytics Portal”.

4.2 Summary of COS Guidelines issued

Table-1: IOM-Summary-COS-Chemistry						
S.N.	IOM/Communication ref. No.	Date of issuance	Issued By	Issued To	Summary	Applicability
1	OS/	20.03.2019	ED(OS)	All HOPs	Uniform Maintenance Work Area Distribution across NTPC stations & JVs (for SWAS reagent procurement)	All NTPC stations and JVs
2	CC/ED(OS)/2019	06.09.2019	ED(OS)	HOPs-All Stations	Chemical Parameters aberration	All NTPC stations and JVs
3	Chlorination plant safe shut- down	25.03.2020	AGM (CO S-Chem)	Station Chemistry heads	Safe shut-down of Chlorination plant in case of Station emergency black-out situation.	All NTPC stations and JVs having chlorination plant
4	OS/OPN/HOD/00/019/12/1	31.03.2020	GM (OS-Operation)	Heads O&M of all coal-based stations	Flame failure at part load	All Coal-based stations of NTPC and JVs
5	Stringent Monitoring of cooling water system parameters	17.09.2020	CGM (COS-Opn, CenPEEP, Comm, Chem)	Heads O&M of all stations	Compliance to CWT online and offline monitoring and availability of online analysers	All NTPC stations and JVs
6	EOC/OS/2020-21/LPDOSING/323002	04.01.2021	GM-COS-C&I	Heads O&M of all stations	LP Dosing automation	All NTPC stations and JVs
7	COS-Chem/Online Monitoring	19.01.2021	CGM (COS-Opn, CenPEEP, Comm, Chem)	Heads O&M of all stations	Online Iron & Turbidity Monitoring	Supercritical and Once-Through Units

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			Chem)			
8	COS-Chem/ SWAS Charging	07.06.2021	GM (COS- Opn, CenPEEP, Comm , Chem)	Heads O&M of all stations	SWAS Charging during unit Start- up activities	All NTPC stations and JVs
9	E-mail communication	29.06.2021	AGM- COS- Chemistry)	Heads of Chemistry	Safe handling of Chemicals	All NTPC stations and JVs
10	E-mail communication	21.07.2021	AGM- COS- Chemistry)	Heads of Chemistry	Iron Monitoring in Feed and MS	All SC units
11	OS-C&I/21-22/site COMM/24	15.02.2022	GM (OS- C&I)	O&M Heads of all Stations with CPU system	Instrumentation for CPU monitoring	All NTPC Stations and JVs with CPU system.
12	COS: OPRN:2022	22.04.2022	ED(OS)	HOPs-All Coal Stations	Guidelines for Import Coal handling & Firing (SOP on Imported Coal Firing)	All Coal- based stations of NTPC and JVs
13	COS-Chem/22- 23/01	28.04.2022	GM (COS- Opn, Com m, Chem)	Heads O&M of all stations	Close monitoring of CEP ACC/DCC/Sodium during unit start- up	All NTPC stations and JVs
14	COS/IOM/ED/22/ 021	06.09.2022	ED (OS)	HOPs of all Stations with CPU system	Streamlining of responsibility related to maintenance aspects of all the jobs related to CPU service vessel area and CPU regeneration area respectively.	All NTPC Stations and JVs with CPU system.
15	COS/IOM/OPN/23/ 03	26.05.2023	GM (COS- Opn, Comm , Chem)	Heads of O&M/ Heads of Chemistry- All 500	Procurement of separate online sodium analyzer for monitoring of main steam samples in 500	All 500 MW drum type unit station

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				MW drum type unit stations	MW drum type units	s (NTPC and JVs)
16	COS/IOM/C&I/22-23/08	24.03.2023	GM (OS-C&I)	O&M Heads of all Stations	HP Dosing pumps AUTO STOP interlock	All 500 MW units
17	COS/IOM/C&I/22-23/07	24.03.2023	GM (OS-C&I)	O&M Heads of all Stations	CPU vessel inlet & outlet valve auto close interlock	All 500 MW units
18	COS/IOM/GM/22	26.07.2023	CGM(OS)	HOPs of VSTPS,FS TPS, SSTPS,Tanda, Unchahar, Kahalgaon, Barauni, Korba, Ramagundam)	Condenser flood test with dye for mixed metallurgy units	Stations having units with mixed metallurgy condenser tubes
19	COS/IOM/ED/24/011	29.02.2024	ED(OS)	HOPs- Stations covered under PAT-VII	PAT-VII Implementation	Stations covered under PAT-VII
20	OS/ST/AGM(OS)/	19.04.2024	AGM (OS-ST)	HOD-TMD- All Coal Stations	Minimum Critical spares for CPU Service Vessel area	All Coal stations

5.0 Water Treatment Plant: Parameter Testing and Operational Records:

5.1 Raw Water/ Clarified water/ Filtered water/Drinking Water

TITLE	Code of Practices on Power Plant Chemistry
LMI No.	LMI/OGN/OPS/CHEM/015 Rev:01

Table-2

S.N.	Parameter	UOM	Raw Water	Raw Water Reservoir (live & Dead)	Clf/Filtered Water	Sand Filter s (I/L & O/L)	Drinking Water/Potable Water
1	pH at 25°C	--	Once/day (Once/four hrs during seasonal variations)	Once/month	Once/day (Once/four hrs during seasonal variations)	---	Once/week
2	Sp. Conductivity at 25°C	μS/cm		Once/month		---	Once/week
3	Turbidity	NTU		Once/month		---	Once/week
4	Total Dissolved Solids (TDS)	mg/l	Once/day	---	Once/day	---	Once/week
5	Ca-Hardness	ppm CaCO ₃	Once/day	---	Once/day	---	Once/week
6	Mg-Hardness	ppm CaCO ₃	Once/day	---	Once/day	---	Once/week
7	Total-Hardness	ppm CaCO ₃	Once/day	---	Once/day	---	Once/week
8	FRC	ppm	Once/day	---	Once/Shift	---	Once/week
9	ORP	mv	Once/day	---	Twice/shift	---	Once/week
10	Sodium	ppm	Once/week	---	Once/week	---	Once/week
11	Potassium	ppm	Once/week	---	Once/week	---	Once/week
12	P-Alkalinity	ppm	Once/day	---	Once/day	---	Once/week
13	M-Alkalinity	ppm	Once/day	---	Once/day	---	Once/week
14	Equivalent Mineral Acidity (EMA)	ppm	Once/day	---	Once/day	---	Once/week
15	Chloride	ppm Cl ⁻	Once/day	---	Once/day	---	Once/week
16	Sulphate	ppm	Once/day	---	Once/day	---	Once/week
17	Reactive Silica	ppm SiO ₂	Once/day	---	Once/day	---	Once/week
18	Colloidal Silica	ppm SiO ₂	Quarterly	---	Quarterly	---	Quarterly
19	Organics (KMnO ₄ No.)	ppm	Once/week	Once/month	Once/week	---	Once/week

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20	Chemical Oxygen Demand (COD)	ppm	Once/week	---	Once/week	---	Once/week
21	Total Suspended Solids (TSS)	ppm	Once/week	---	Once/week	---	Once/week
22	Total Iron	ppm Fe	Once/week	---	Once/week	---	Once/week
23	Total Organic Carbon (TOC)	ppm	Once/week	---	Once/week	Quarterly	Once/week
24	Biochemical Oxygen Demand (BOD)	ppm	Once/fortnight	---	Once/fortnight	---	Once/fortnight
25	Oil & Grease	ppm	Once/month	---	Once/month	---	Once/month
26	Heavy Metal Analysis	---	Once/Year (External-NETRA or CSIR/Govt/ NABL Lab)	---	Once/Year (External-NETRA or CSIR/Govt/ NABL Lab)	---	Once/6 months (External-NETRA or CSIR/Govt/ NABL Lab)
27	Jar test for raw water only (with different coagulant concentrations)	---	Once/month	---	---	---	
28	All parameters as per IS 10500-latest version (drinking water)	---	---	---	---	---	Once/6 months (External-NETRA or CSIR/Govt/ NABL Lab)

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Note: Refer OIN-OPS-CHEM-017 on “Manual on Analytical Test Procedures for Water and Steam in Power Plants” for water parameter test methods.

5.2 Guidelines for Operation and Maintenance of Resin based DM Plant

Online Instrumentation for Monitoring the DM Plant: Refer table 13B

Guidelines for Critical Valves in DM Streams:

- The acid and caustic dosing pneumatic valves for the SAC, SBA, and MB vessels should be of the air-to-open type.
- A pneumatic bleed valve should be available in the acid and caustic dosing lines of all regeneration vessels.
- The bleed valve should be of the air-to-close type.

Important Guidelines for the Operation of DM Stream:

- Free residual chlorine (FRC) in the range of 0.10-0.20 ppm should be maintained in the filter water sump (sample point: filter water pump common discharge header) when Chlorine, Hypo solution, or bleaching powder is used as a biocide in the pre-treatment process. The FRC at the ACF outlet must be maintained at 0.0 ppm without any deviation.
- The DM streams should be operated in automatic or semi-automatic mode, and all pneumatic valves should be operated from OWS only.
- In case DM stream is operated in manual mode (as per design), then provision needs to be made for independent hooter alarm (on high MB outlet specific conductivity) in control room for manual tripping of DM Plant.
- DM streams should always be operated at the design flow rate.
- DM stream load projection should be carried out based on actual raw water analysis on daily basis.
- The DM stream should be operated based on the following recommended chemistry parameters:

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Table-3A:

Sample	Sodium (ppm), max	Conductivity comparator ratio, max	CO₂, ppm, max	Sp. Cond (uS/cm) at 25°C, max	Silica (ppb), max	Frequency
SAC	2.0	1.1	--	---	--	
Degassed water	--	--	5.0	---	--	Once/shift
SBA	--	--	--	10	200	Twice/shift
MB	--	--	--	0.10	20	Twice/shift

Note-

- i. Except for CO₂, all values should be monitored using online analysers. Offline analysis should be conducted according to the frequency mentioned table 3B and 3C.
- ii. Inspection of degasser tower and performance of degasser blower are to be checked if degassed water free CO₂ is continuously maintained >5.0 ppm.
- iii. Over run of DM streams should not be done.

- The online parameters of the DM stream, such as flow, DP, conductivity, sodium, silica, etc., should be recorded in the DM plant logbook.
- DM streams cross operation should be avoided unless necessary. Site should ensure the tripping of service-in ion exchanger vessels on high conductivity of individual MB & their common header during cross operation.

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S. N.	Parameter s/ Checks	UOM	ACF Outlet	Cation Exchanger	Degasser	Anion Exchanger	Mixed Bed (Individual Stream)	Mixed Bed (Common Header)	DM Storage tank	DM Make-up pump (suction/ discharge header)
-------	---------------------	-----	------------	------------------	----------	-----------------	-------------------------------	---------------------------	-----------------	---

Table-3B: Parameters to be monitored for Resin-Based DM Plant:

1	pH at 25 deg C	---	---	---	---	Online: Continuous, Offline: Twice/ shift	Online: Continuous, Offline: Twice/ shift	---	Online: Continuous, Offline: Once/D ay	---
2	Sp Conductivity at 25 °C	µS/cm	---	---	---	Online: Continuous, Offline: Twice/ shift	Online: Continuous, Offline: Twice/ shift	Online: Continuous, Offline: Twice/ shift	Online: Continuous, Offline: Once/D ay (Make-up water line)	Online: Continuous
3	Turbidity	NTU	Once / shift	---	---	---	---	---	---	---
4	FRC	ppm	Once / shift	---	---	---	---	---	---	---
5	Sodium	ppm	---	Stage-II	---	---	---	---	---	---
6	Conductivity Ratio		---	1 Stage-1	---	---	---	---	---	---
7	FMA	ppm	---	Once / shift	---	---	---	---	---	---
8	Free CO2	ppm	---	---	Once / shift	---	---	---	---	---
9	Blower Discharge Pressure	mm WC	---	---	Once / shift	---	---	---	---	---

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10	Reactive Silica	ppm as SiO ₂	---	---	---	Online: Continuous, Offline: Twice/shift	Online: Continuous, Offline: Twice/shift	Online: Continuous, Offline: Twice/shift	Online: Continuous, Offline: Once/Day	---
11	Colloidal Silica	ppm as SiO ₂	---	---	---	---	Offline-Quarterly	---	Offline-Quarterly	---
12	OBR (Totalised flow)	M3	Once/Shift	Once/Shift		Once/Shift	Once/Shift	---	---	---

Table-3C: Protection-Interlock Checks

1	DM Stream Tripping/Alarm (Auto operation)						Tripping at 0.1 μ S/cm or Design Value	Tripping at 0.5 μ S/cm		Alarm at 2.0 μ S/cm (make-up water line)
---	---	--	--	--	--	--	--	----------------------------	--	--

Important Guidelines for regeneration of DM streams:

- DM plant regeneration should preferably be performed in automatic or semi-automatic mode to avoid deviations in the regeneration process.
- The service outlet manual valve of idle MB should be kept closed during regeneration of Cation or Anion ion-exchange vessels.
- The bleed valves should be fully opened automatically during the service and idle modes of any ion exchange vessel.
- The root valve of online analysers should be closed during regeneration process (except fast rinse).
- The recommended power water flow should be ensured before starting the chemical injection process.
- Propper separation of MB resin with extended controlled backwash must be ensured before starting chemical injection process.
- The recommended cation level be ensured in MB to avoid improper regeneration and poor effluent water quality.
- HCl and NaOH (conc. as per design) should be used for the regeneration of strong cation and anion resins, respectively. Hot water should always be used for the regeneration of anions in WBA, SBA, and MB vessels.
- Design quantity of HCl and NaOH should be used for regeneration.
- Approx 50-100mm water level to be maintained in MB for proper mixing of cation and resins.

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- k. Vent valve of ion exchanger vessels should not be opened during their regeneration and rinsing.
- l. The power water flow, acid and caustic concentrations, injection time, slow rinse time, fast rinse time, etc., should be recorded in the DM regeneration logbook.
- m. Recording rinse times provides an early indication of resin condition, as extended rinse times indicate resin fouling.
- n. A separate logbook should be maintained to record regeneration activities.
- o. Proper RCA is required for regeneration failure, if occurred. Record is to be maintained.
- p. The healthiness of all dosing pumps and ejectors should be ensured.
- q. A check list is to be used for regeneration and open backwash of ACF, Cation, Anion and MB ion-exchanger.

Recommended Maintenance Guidelines for DM Plant:

- a. SBA resin fouling test to be carried out once/year and record is to be maintained.
- b. SBA brine treatment record is to be maintained.
- c. Annual quality analysis of ACF and ion exchangers resin are to be carried out as specified below:
- d. Immediate replacement of ACF to be carried out based on the latest iodine value report.
- e. The recommended inspection schedule is to be followed for a healthy DM plant.

Recommended Critical Spares for DM plant:

- a. ACF and resins- Responsibility: Site Chemistry
- b. COS-BOP guidelines to be followed- Responsibility: Site MMD-OFS.
- c. Spares for pneumatic valves, conductivity analyser etc- Responsibility: Site C&I maintenance.

• Common Guidelines for resin based DM plant:

- a) The DM streams should be operated in automatic or semi-automatic mode, and all pneumatic valves should be operated from OWS only.
- b) Six-monthly checks of all protections and interlocks must be conducted. A joint protocol should be maintained by the site Chemistry and C&I.
- c) Six-monthly checks for tripping of the DM stream at high specific conductivity (individual MB outlet and common header) must be ensured, with a joint protocol maintained by Chemistry and C&I.
- d) A register should be maintained to record protection interlock bypass details.
- e) DM storage tank filling and consumption **should never be done** simultaneously in the same tank. DMST filling and make-up done from/ to which tank to be noted in shift log book.
- f) Fortnightly rinse of ACF and ion exchangers should be ensured during idle period, with minimum two to three resin bed volume of water.
- g) Resin level in all ion-exchanger vessels should be recorded fortnightly and top-up of

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resins to be done, if required. A record is to be maintained.

- h) Changeover schedule of critical equipments of PT and DM plant is to be recorded.
- i) Equipment non-availability > 15 days (template as per annexure-I).

5.3 WATER PRODUCTION/ CONSUMPTION DATA

Table 4:

S.N	Type of water	Opening stock (M3)	Production (M3)	Consumption (M3)	Closing Stock (M3)	Previous Month consumption (M3)
1	Clarified Water					
2	DM Water (Cyclic & Non-Cyclic)					

5.4 DM WATER AND CLARIFIED WATER COST CALCULATIONS

Site shall maintain monthly records of DM water and clarified water cost calculations. An illustrative Template for cost calculation is enclosed as Annexure-II.

5.5 DM PLANT PERFORMANCE DATA (MONTHLY UPDATION)

Table 5:

No of regenerations of cation exchanger	No of regenerations of anion exchanger	No. of regeneration of MB	Qty of resin top up (indicate vessel and name of the resin)	Quantity of DM water consumed as make up	Quantity of DM water consumed for regeneration	Brine treatment, if any (acid/ brine+ caustic)

5.6 Bulk Chemical Consumption

Table 6A:

S.N.	Name of Chemical	SAP Mtl Code	UOM	Opening Stock	Qty. Recd	Total Consumption	Final stock	REMARKS (Any additional information)
------	------------------	--------------	-----	---------------	-----------	-------------------	-------------	--------------------------------------

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1	Ferric Alum		MT					
2	Hydrated Lime		MT					
3	Hydrochloric Acid (30%)		MT					
4	Caustic Soda Lye(100%)		MT					
5	Caustic Soda Flakes		MT					
6	Liq.Ammonia		L					
7	Bleaching Powder		MT					
8	Chlorine		MT					

NOTES:

- i. Any other site-specific bulk chemicals (eg. Remineralization chemicals) maybe included in the list in table 8A.
- ii. Safety instructions and relevant MSDS for all bulk chemicals shall be under display at key locations of WTP and CWT area.
- iii. OIN-OPS-CHEM-011 on “Analytical Procedures for Raw Materials” may be referred for test methods of bulk chemical quality checks.

5.7 Specific Chemical Consumption Data for DM Plant**Table 6B:**

S.N.	Name of the chemical	Design Specific Consumption /As per existing ionic load (kg/m3)	Actual (Stage wise) (kg/m3)	Station Sp. Consum ption (kg/m3)	Sp. Consum ption of previous month	Targe t (finan cial year)
1	HCl					
2	NaOH					

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5.8 Activated Carbon/ Resin Testing (DM Plant, Softening Plant, CPU)

Table 7A: Resin testing				
S.N.	Resin type	Parameters to be tested	Testing Frequency	Remarks
1.	WAC (Weak Acid Cation)	- Total Exchange Capacity - Water Retention Capacity - Density	Once/Year	Activated carbon and Resins testing to be done at CC-NETRA/ Govt./NABL-accredited third party labs/ ACF/Resin supplier's lab.
2.	SAC (Strong Acid Cation)	- Total Exchange Capacity - Salt Splitting Capacity - Water Retention Capacity - Density	Once/Year	
3.	WBA (Weak Base Anion)	- Total Exchange Capacity - Water Retention Capacity - Density	Once/Year	
4.	SBA (Strong Base Anion)	- Total Exchange Capacity - Salt Splitting Capacity - Water Retention Capacity - Density	Once/Year	
5.	ACTIVATED CARBON	Iodine value	Once/Year	

Note: Following parameters are to be tested for Cation/Anion resins in case station is having problems with OBR reduction:

- i. Operating Capacity
- ii. Particle size distribution

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Table 7B: Suggested minimum stock of DM plant resins						
1. DMP Resin type (no. of lots)- For plants having 02 no's of DM streams						
WAC resin (No. of lot)	WBA resin (No. of lot)	SAC resin (No. of lot)	SBA resin (No. of lot)	ACF resin (No. of lot)	MB Resin	Remarks
2	2	1	1	1	1 lot each of Cation & Anion	Resin replacement programme to be made in case of >20% capacity reduction.
2. DMP Resin type (no. of lots)- For plants having 03 no's of DM streams						
3	3	2	2	2	2 lots each of Cation & Anion	Resin replacement programme to be made in case of >20% capacity reduction.
2. CPU resins	Refer OGN-OPS-CHEM-020 on "Guidelines on Operating Philosophy of Condensate Polishing Unit)					

5.9 SAMPLE LOG SHEETS

A.DM PLANT LOG SHEET

Station:

Date:

DATE	TIME	FWTP		ACF		CATION		DEGASSER					ANION		MB	
		A / B / C		A / B		A / B		PUM P No.		Blower PRE. Kg/Cm2	TAN K LEVE L - MTR		A / B		A / B	
		FLO W	PRE. Kg/C	FLO W	PRE. Kg/C	FLO W	PRE. Kg/C	A/B/C	PRE. Kg/C		A	B	FLO W	PRE. Kg/C	FLO W	PRE. Kg/C

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6.0 Main Plant: Parameters to be tested/ recorded for Steam Water Cycle and their normal frequency of testing/recording:

Table 8A:					
S.N.	PARAMETER	UOM	Online	Offline	Remarks
			Test frequency	Test frequency	
A	MAKE-UP WATER (Unit Side)				
1	Sp. Conductivity at 25°C	µs/cm	Continuous	Once/shift	
2	ACC	µs/cm	Continuous	Once/week	
3	DCC	µs/cm	Continuous	----	
4	Silica, reactive	ppb	Continuous	Once/shift	
B	CONDENSATE				
1	pH at 25°C	----	Continuous	Once/shift	
2	Sp. Conductivity at 25°C	µs/cm	Continuous	Once/shift	
3	ACC	µs/cm	Continuous	Once/week	
4	DCC	µs/cm	Continuous	----	
5	Silica, reactive	ppb	Continuous	Once/shift	
6	Ammonia	ppb	Continuous	Once/shift	
7	Dissolved O2	ppb	Continuous	Once/week	
8	Total Iron (dissolved & suspended)	ppb	-----	Once/day	
9	Sodium	ppb	Continuous	----	
10	CRUD	ppb	-----	Once/week	Sample volume: 100 L (Minimum)
11	Turbidity	NTU	Continuous	----	
C	FEED WATER (Economizer Inlet)				
1	pH at 25°C	----	Continuous	Once/shift	
2	Sp. Conductivity at 25°C	µs/cm	Continuous	Once/shift	
3	ACC	µs/cm	Continuous	Once/week	
4	DCC	µs/cm	Continuous	----	
5	Silica, reactive	ppb	Continuous	Once/shift	
6	Dissolved O2	ppb	Continuous	Once/week	
7	CRUD	ppb	-----	Once/day	Sample volume: 100 L (minimum)
8	Sodium	ppb	Continuous	-----	

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D.	BOILER WATER				
1	pH at 25°C	----	Continuous	Once/shift	
2	Specific Conductivity at 25°C	µs/cm	Continuous	Once/shift	
3	Drum ACC	µs/cm	Continuous	Once/week	
4	Silica, reactive	ppb	Continuous	Once/shift	
E.	SEPARATOR OUTLET/ SATURATED STEAM				
1	pH at 25°C	---	Continuous	Once/shift	
2	Specific Conductivity at 25°C	µs/cm	Continuous	Once/shift	
3	ACC	µs/cm	Continuous	Once/week	
4	Silica, reactive	ppb	Continuous	Once/shift	
5	Sodium	ppb	Continuous	-----	
F.	MAIN STEAM				
1	pH at 25°C	---	Continuous	Once/shift	
2	Specific Conductivity at 25°C	µs/cm	Continuous	Once/shift	
3	ACC	µs/cm	Continuous	Once/week	
4	Silica, reactive	ppb	Continuous	Once/shift	
5	Sodium	ppb	Continuous	-----	
6	Ammonia	ppb	-----	Once/shift	
7	Total Iron	ppb	-----	Once/day	
8	CRUD	ppb	-----	Once/week	Sample volume: 100 L (minimum)
9	Chloride	ppb	-----	Once/week	
10	TOC	ppb	-----	Once/week	
Note: ACC is Cation Conductivity/After Cation Conductivity.					

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Table 8B:

S.N.	PARAMETER	UOM	Online	Offline	Remarks
			Test frequency	Test frequency	
G	ECW (TG&SG)				
1	pH	---	Continuous	Once/day (unitwise)	
H	Condensate storage tank				
1	pH		-----	Once/week	
2	Reactive Silica	ppm	-----	Once/week	
3	Specific conductivity	µs/cm	-----	Once/week	
4	Total Iron	ppb	-----	Once/week	
5	Chloride	ppb		Once/week	
I	CBD Opening for drum units (daily operation)				
1	CBD Opening Hours (Unit wise)	Hrs	-----	-----	Reasons to be recorded
J	Generator Stator Primary Water (parameters)	Refer OGN-OPS-CHEM-002 (latest version) on “Chemical Control of Generator Stator Primary Water System”			
K	Condensate Polisher Unit (Parameters)	Refer OGN-OPS-CHEM-020 (latest version) on “Guidelines on Operating Philosophy of Condensate Polishing Unit”.			

Table 8C: SWAS online instrument availability record

Table 001: Online Instrument Availability Record							
Name of the station:				Stage:	Unit:		
Month:				Date:			
S.N.	Name of the Analyser	Availabil ity (Yes/No)	Reason for non- availabili ty	Last calibrati on date	Online/Offline value comparison		Remarks
					Online	Offline	

Note: Joint protocol is to be maintained by site Chemistry and C&I.

Frequency: Weekly.

- Data to be recorded unit wise and limits for various parameters are to be specified as per latest version of OD/OPS/CHEM/001 on "Chemical Control of the Water-Steam Circuits of Drum & Once- Through Type Boilers and HRSG").
- Max, min, and average values; Number of samples outside the prescribed limits are to be recorded.
- Reasons for persistent deviation are to be recorded. Root Cause Analysis of chemistry parameter aberrations is to be done on regular basis based on OPMS data.

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7.0 Cooling Water System

Monitoring parameters, templates and limits shall be as per **OGN-OPS-CHEM-019** (latest version) on "Guidelines on Cooling Water Treatment, Monitoring and Chemical Cleaning of Condenser Tubes".

8.0 Coal & Bio-fuel Analysis

A. COAL-AS-RECEIVED- INDIGENOUS COAL

Table 9A

Year: Month:

Name of the Mines/Siding:

Testing frequency: As and when received

Receiving Frequency: As and when received										
Date of receipt	RR No	RR Date	Quantity (MT)	Total moisture (%)	Equilibrated Moisture Basis (40°C & 60% RH)				Grade	Remarks
					Moisture	Ash	VM	GCV		
					(%)	(%)	(%)	(Kcal/kg)		
Design Limits/ As per FSA										

Monthly weighted average GCV:

NOTE:

- Site may append additional columns, if required. But the above details shall be included in the template mandatorily.
- Modalities of third party sampling and FSA may be taken into account for test report preparation.
- Loading end samples to be tested and records maintained as per requirements of CIMFR third party sampling/FSA/any other statutory requirements.
- SAP data updation for test parameters of unloading end rakes is to be done on daily basis.

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- v. Siding wise Sulphur data is to be maintained by individual sites.
- vi. Monthly weighted average GCV values (and all other parameters if required) to be calculated based on quantity of coal received.
- vii. OGN-OPS-SYST-048 on "Guidelines on Supply of Domestic and Imported Coal" and guidelines of CC-FM shall be applicable.

B. Coal-as-Received- Imported Coal

Table 9B

Year: Month:

Name of the Mines/Siding:

Testing frequency: As and when received

Vessel name	Date of receipt	Date of sampling					QTY (MT)	Total	Size	Inherent	Ash (%)	Fixed carbon	GCV	Sulphur (%)	REMARKS
								As-received basis	Air-Dried Basis & TM basis						

NOTE:

- i. Site may append additional columns, if required. But the above details are to be included in the template mandatorily.
- ii. Modalities of imported coal bidding documents/FSA are to be taken into account for test report preparation.
- iii. Site shall keep records of supplier's test reports of load port.
- iv. Any other parameter as per imported coal agreement shall be appended by station, as and when required.
- v. Monthly weighted average GCV values (and all other parameters if required) to be calculated based on quantity of coal received.
- vi. Guidelines of IOM ref. No. COS:OPRN:2022 dated 22.04.2022 is to be ensured.

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- vii. OGN-OPS-SYST-048 on “Guidelines on Supply of Domestic and Imported Coal” and guidelines of CC- FM shall be applicable.

C. Coal As-fired

Table 9C

Testing frequency: Daily basis/Stagewise

Proximate Analysis: Bulk/Stage-wise								
S.N.	Date of sampling	Qty (MT)	Total Moisture (%)	Ash	VM	FC	GCV	Remarks
				(%)	(%)	(%)	(Kcal/kg)	
				(TM basis)				

NOTE:

- Site may append additional columns, if required. But the above details shall be included in the template mandatorily.
- Site may carry out unit wise as-fired coal analysis if the station is facing any specific problems related to quality.
- Monthly weighted average values shall be calculated based on quantity of coal received.
- Guidelines on imported coal handling and firing and associated SOP (IOM ref. No. COS:OPRN:2022 dated 22.04.2022) is to be ensured.
- Volatile Matter (%) of feeder coal is to be monitored on daily basis [refer IOM issued by GM (OS-Opn) ref.no. OS/OPN/HOD/00/019/12/1 dated 31.03.2020].

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D. Ultimate Analysis & Ash Analysis**Table 9D**

S.N.	Analysis	Frequency	Remarks
1.	ULTIMATE ANALYSIS ▪ CARBON ▪ HYDROGEN ▪ NITROGEN ▪ SULPHUR ▪ OXYGEN ▪ PHOSPHORUS ▪ CHLORINE ▪ HEAVY METALS (Hg, Pb, As)	Once/6-months	If testing facilities are not available at site, sample are to be sent to CC-NETRA/CSIR labs/Govt/ NABL accredited labs.
2.	ASH ANALYSIS (Complete chemical characteristics including alpha-quartz)	Once/year (Once/six months in case of clinkering issues)	---do--
3.	ASH FUSION TEMPERATURE ▪ INITIAL DEFORMATION ▪ SOFTENING ▪ HEMISPHERICAL ▪ FLUID	Once/year (Once/six months in case of clinkering issues)	---do---
4.	HARDGROVE GRINDABILITY INDEX (HGI)	Once/Month or as per requirement	--do--
5.	ABRASION INDEX (YGP INDEX)	Once/6-months	--do--

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E. Mill Fineness

Testing Frequency: All mills in service in each unit should be covered at least once in a month.

Table 9 F

S. N	Date	Unit	Mill	Load	Coal Flow	Air Flow	PA Fan Comm on Header Pressure	Mill I/L Temp	Mill I/L Temp	Burner Tilt	Mill Current	% Coal retained on 50 mesh sieve (300 micron IS sieve)	% Coal passing through 100 mesh sieve (150 micron IS sieve)	% Coal passing through 200 mesh sieve (75 micron IS sieve)
				(MW)	(T/hr)	(T/hr)	(mm WC)	(deg C)	(deg C)	(%)	(A)			
LIMITS														
MONTHLY DATA	AVERAGE													
	MAXIMUM (DATE)													
	MINIMUM (DATE)													
	No of Deviations													
REMARKS:														

Note: Isokinetic sampling for mill fineness is to be done for each mill at least once in six months or as per site-specific requirement, in coordination with site EEMG. Site EEMG shall arrange for isokinetic sample collection with their own equipment, mill fineness test shall be done by site Chemistry and results are to be compared with regular mill fineness data.

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F. Mill Rejects (Dry basis) Unit wise/Stagewise (as the case may be)

Testing Frequency: At least all mills in service per unit should be covered in a month/As per site requirement. Samples to be provided by Site BMD.

Table 9G					
S.N.	Date	Unit/ Stage	Ash (%)	GCV (Gross Calorific Value) (Kcal/kg)	Remarks

G. Combustibles in Bottom Ash and Fly Ash (Dry Basis)

Testing Frequency: Combustibles in BA-FA to be monitored daily for all running units. If imported coal is being blended, then ShiftWise sampling and analysis is to be done (Ref IOM COS: OPRN:2022 dated 22.04.2022).

Table 9H								
S.N.	Month	Stage	Unit	Date	Mills in service	% Combustibles		Remarks
						Bottom Ash	Fly Ash	
LIMITS (OEM recommendation)								
MONTHLY DATA			AVERAGE					
			MAX (DATE)					
			MIN (DATE)					
			No of Deviations					

H. Coal Stock-pile GCV mapping & bulk density measurement

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Testing Frequency: Monthly once (first week of the month preferably)

Table 9I			
S.N.	Coal Stock Pile ID	GCV (kcal/kg)	Bulk Density
Note: Samples collected at Different Stack levels is to be mentioned.			

I. Biomass pellets analysis

Table 9J							
S.N.	Sample ID	Date of Sampling	Fines (< 3 mm)	Size Distribution (%)		As-Received Basis	
			ARB Wt %	1 mm	3 mm	TM %	GCV (kcal/kg)

9.0 Limestone & Gypsum Analysis for FGD

Table 9K: Chemical Parameters for Limestone								
S.N.	Sample ID	Date of Sampling	LOI (At 800 deg C)	CaO (%)	MgO (%)	SiO₂ (%)	Fe₂O₃ (%)	Al₂O₃ (%)

Table 9L: Chemical Parameters for Gypsum						
S.N.	Sample ID	Date of Sampling	Moisture (%)	Purity (%)	SiO₂ (%)	Chloride (ppm)

TITLE	Code of Practices on Power Plant Chemistry
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10.0 Oil Analysis

A. Fuel Oil Analysis

Testing Frequency: As and when received

Table 10A		
Name of the station:		Stage:
Date of sampling	Parameters	
	Density	Moisture
UOM	g/ml	% w/w

B. Lube oil and Control Fluid/FRF Analysis

Table. 10B								
Name of the Station:				Stage:				
Month:				Date of sampling				
S.N .	Type of Oil	Parameters to be tested						
		Moisture	Kinema tic Viscosi ty	Neutralisat ion number	Mecha nical Impurit ies	Particle Distribut ion (NAS- 1638/IS O- 4406)	PQ Index	Remarks
		ppm	cst	mg KOH/g oil	ppm			
Test Frequency								
1	MOT	Weekly	Weekl y	Weekly	Weekly	Monthly	----	Six-monthly external testing from IOCL/supplier /CC- NETRA
2	MDBFP	Weekly	Weekly	Weekly	Weekly	Monthly		
3	TDBFP	Weekly	Weekly	Weekly	Weekly	Monthly		
4	FRF	Fortnightly	Monthl y	Fortnightl y	---- -	Monthly	----	Six-monthly external testing from CC- NETRA

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5	LP-HP Bypass Oil (FRF)	Monthly	Quarterly	Monthly	----	Monthly	----	Yearly once external testing from CC-NETRA/OEM/supplier
6	LP-HP Bypass Oil (lub oil)	Quarterly	Quarterly	Quarterly	Quarterly	Quarterly	----	
7	Fan Oil	As per OGN-OPS-MECH-013 and OGN-OPS-MECH-051						
8	Mill Gear Box Lub Oil	Quarterly	Quarterly	Quarterly	Quarterly	----	Quarterly	
9	Mill Hydraulic Oil	Quarterly				Quarterly		
10	APH Oil	As per OGN-OPS-MECH-055						
11	CW Discharge Butterfly valve-hydraulic oil	Once/Six months	----	----	Once/Six months	----	----	
12	CHP Oils							
12a. Hydraulic Oil (Contamination Analysis)								
a	Wagon Tippler	----	----	----	----	Monthly	Monthly	
b	Paddle Feeder	----	----	----	----	Monthly	Monthly	
c	Stacker Reclaimer	----	----	----	----	Monthly	Monthly	
12b. Gear Box Oil								
a	Coal Crusher (Ring Granulator) Scoop-type Fluid Coupling	Quarterly	Quarterly	----	----	----	Quarterly	
b	Stacker-Reclaimer Bucket Wheel Gear Box	Quarterly	Quarterly	----	----	----	Quarterly	
c	Wagon Tippler Main Drive Gear Box	Quarterly	Quarterly	----	----	----	Quarterly	
d	Apron Feeder Gear Box	Quarterly	Quarterly	----	----	----	Quarterly	

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			y				rly	
e	Vibro Feeder Gear Box	Quarterly	Quarterly	-----	-----	-----	Quarterly	
f	Bunker Conveyor	Quarterly	Quarterly				Quarterly	

General Guideline Notes for Oil Sampling & Analysis:

- MOT and FRF oil samples shall be collected in presence of Operation Department representative. Fan oil/APH oil/Mill oil and CHP oils shall be provided to Chemistry lab by BMD and CHP representatives respectively (as per the test frequency).
- HP & LP bypass oil analysis- as per site specific requirement.
- During sample collection proper flushing of the sample line and draining of residual oil in sample line is to be ensured so that there is no external contamination in the final sample which is collected for testing of parameters.
- All oil samples are to be visually examined for appearance (if emulsified or particulates present).
- For acceptance criteria and limits of MOT and FRF, and for responsibility matrix, refer latest versions of OGN-OPS-CHEM-021 (on "The Examination and Maintenance of Lubricating Oils of Turbine and Boiler Feed Pumps") and OGN/OPS/MECH/001 (on "The Examination, Maintenance, Safety & Handling Aspects of Phosphate Ester Fire Resistant Hydraulic Fluids) or as per OEM recommendations.
- For testing parameters related to oil top-up and replacement refer latest versions of OGN-OPS-MECH-001 (on "The Examination, Maintenance, Safety & Handling Aspects of Phosphate Ester Fire Resistant Hydraulic Fluids) and OGN-OS-CHEM-021 (on "The Examination and Maintenance of Lubricating Oils of Turbine and Boiler Feed Pumps").
- For CHP Hydraulic oils, recommendations of OGN-OPS-SYST-057_Rev0_Reliability Improvement Matrix of Coal Handling Plant (latest version) are applicable.
- For CHP gear box oil, recommendations of IOM: OS/BoP/IOM/20-21/20 Dated 25.11.2021 are applicable.
- OIN-OPS-CHEM-015 on "Manual on Sampling and Testing Procedures for Oils" may be referred for
test procedures of oil monitoring parameters.
- Whenever oil samples are being sent for external testing, station should ensure testing of the same sample in site laboratory as per available facilities.
- In case of any abnormality/increasing trend observed in any oil sample w.r.t. PQI value, site (in consultation with the maintenance counterpart) may please send a sample to CC-NETRA for complete Wear Debris Analysis.

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C. Transformer Oil Dissolved Gas Analysis

Table 10C

**Date of
sampling:**

Date of Analysis:

**Testing
Technique:**

Transformer no:

Oil Type:

Instrument:

PARAMETERS	H2	CH4	CO	CO2	C2H4	C2H6	C2H2
UOM	PPM	PPM	PPM	PPM	PPM	PPM	PPM
Limits							
ANALYSIS RESULTS							
SAMPLE IDs							

Note:

- DGA results are to be maintained on COS website online portal "Tx Monitoring"- "DGA Monitoring Application".
- For frequency of DGA, guidelines of OGN-OPS-ELEC-038 on "Compendium for Maintenance Guideline for EHV Transformers for enhanced reliability" (latest version) shall be applicable.

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11.0 Gas Analysis

A. Flue gas (by ORSAT or portable gas analyzer)

Table 11A				
Name of the Station:		Stage:		
Month:				
S.N.	Date of Sampling	Parameters (%)	Test frequency	Remarks
1		CARBON DIOXIDE	Once/week/unit.	Online CO2/ O2 analysers to be cross-checked by ORSAT/Portable Gas Analyser.
2		OXYGEN		Flue gas analysis to be done at PAPH/SAPH outlet and eco outlet.

B. Hydrogen Gas

Table 11B				
Name of the Station:		Stage:		
Month:				
S.N.	Date of Sampling	PARAMETERS	TEST FREQUENCY	REMARKS
1		H2 purity (generator)	Online continuous, offline once/day	By ORSAT/Portable H2 purity analyser
2		Dew point (generator H2)	Online continuous, offline once/week	By Dew Point Apparatus
3		H2 purity (H2 cylinder)	As per site requirement	By ORSAT/Portable H2 purity analyser.

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C. Instrument Air

Table 11C				
S.N.	Date of Sampling	Parameters	Test frequency	Remarks
1		Dew point- Instrument Air Compressor (IAC)- Drier Outlet and Receiver Tank Outlet	Online continuous, offline once/week.	By Dew Point Apparatus Air sample to be taken from body of receiver tank.
2		Dew point-Transport Air Compressor (TAC) of Ash Handling Plant	Online continuous, offline once/fortnight.	By Dew Point Apparatus

12.0 Laboratory Equipments

A. Water Analysis Instruments

Table 12A	
S.N.	NAME OF THE EQUIPMENT
1	Atomic Absorption Spectrophotometer (AAS) #
2	UV-VIS Spectrophotometer (Single beam & Double beam)
3	Ion Chromatograph
4	Flame Photometer
5	Turbidity Meter (Nephelometric)- Benchtop & Portable
6	Dissolved Oxygen Analyser (Optical Sensor based)
7	Conductivity meters (Bench top & Portable- Both high range & low range)
8	Jar test apparatus (with RPM regular and lighted background screen)
9	pH Meter (Bench top and portable)
10	ORP Meter (Bench top and portable)
11	WEIGHING BALANCE, single pan

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(a)	0-200 g (LC:0.1mg)
(b)	0-200 g (LC: 0.01 mg)
12	Portable offline ACC Meter
13(a)	CRUD Testing Kit- Vacuum flask-type
13(b)	In-line Stainless Steel Filter Holder (for CRUD testing)- For quantitative analysis
14	ICP-OES & Colloidal Silica digester (optional, based on site requirement) #
15	Total Organic Carbon Analyzer
16	Ultrapure Water System

B. Coal Analysis Instruments

Table 12B

S.N.	NAME OF THE EQUIPMENT
1	AUTOMATIC BOMB CALORIMETER
2	WEIGHING BALANCE, ANALYTICAL, SINGLE PAN (LC: 0.1 mg or better)
3	WEIGHING BALANCE, TOP LOADING (digital)
4	COAL SAMPLE RIFFLE
5	FURNACE FOR VM DETERMINATION
6	MUFFLE FURNACE
7	HGI ANALYSER
8	HUMIDITY OVEN FOR EQUILIBRATED MOISTURE
9	JAW CRUSHER & DOUBLE ROLL CRUSHER
10	RAYMOND MINI MILL
11	COAL PULVERISER
12	SIEVE SHAKER WITH SET OF STANDARD SIEVES
13	PROXIMATE ANALYSER, AUTOMATIC
14	LABORATORY HOT-AIR OVEN
15	CYCLONE SEPARATOR
16	SULPHUR ANALYSER

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C. Oil Analysis Instruments

Table 12C

S.N.	NAME OF THE EQUIPMENT
1	FLASH POINT APPARATUS (automatic)
2	VISCOMETER (Kinematic) for all ranges with viscometer bath
3	VISCOMETER (Saybolt/Redwood) [#]
4	AUTOMATIC KARL FISCHER TITRATOR: Coulometric (for low range) & Volumetric (for high range)
5	PQ Index (FERROUS INDEX) MONITOR
6	DEAN & STARK APPARATUS ALONG WITH HEATING MANTLE OF 1 LITRE FLASK CAPACITY
7	LABORATORY CENTRIFUGE MACHINE
8	MILLIPORE FILTER ASSEMBLY
9	0.45 AND 0.8 MICRON SIZE FILTER PAPERS FOR MILLIPORE FILTER ASSEMBLY
10	ROTARY VACUUM PUMP DOUBLE STAGE
11	MAGNETIC STIRRER
12	DISSOLVED GAS ANALYZER (complete with extraction/head-space sampler and analyser assembly)
13	OIL PARTICLE COUNTER (NAS Analyzer)
14	THERMOSTAT BATH

D. Instruments for Flue Gas, DGA and Hydrogen Purity

Table 12D

S.N.	NAME OF THE EQUIPMENT
-------------	------------------------------

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1	ORSAT APPARATUS (Burette Least Count should be 0.05 ml or better)
2	MULTI GAS DETECTOR
3	PORTABLE FLUE GAS ANALYZER
4	DEW POINT METER
5	HYDROGEN PURITY METER

E. Cooling Water Analysis Instruments

Table 12E	
S.N.	NAME OF THE EQUIPMENT
1	DEPOSIT MONITER (As per NACE standard)
2	BIOFOULING METER (As per NACE standard)
3	CORRATOR (As per NACE standard)
4	CORROSION RACK
5	CORROSION COUPONS (depending upon metallurgy): MS/SS/Cupro Nickel (Cu-Ni)/Titanium

F. Environmental Monitoring Instruments

Table 12F	
S.N.	NAME OF THE EQUIPMENT
1	STACK MONITORING KIT
2	RESPIRABLE DUST SAMPLER
3	BOD ANALYSER (With incubator) #
4	COD ANALYSER#

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G. General Laboratory Instruments/Equipments.

Table 12G

S.N.	NAME OF THE EQUIPMENT
1	BALANCE ANALYTICAL, SINGLE PAN (LC: 0.1 mg & 0.01mg)
2	DENSITY HYDROMETERS (All Ranges as per site requirement)
3	HYGROMETER (Direct Reading Type- Portable)
4	ANEROID BAROMETER
5	HOT PLATE
6	PESTLE AND MORTAR (Ceramic)
7	PSYCHROMETER
8	HOT PLATE CUM MAGNETIC STIRRER
9	MEMBRANE FILTER ASSEMBLY
10	LABORATORY OVEN FOR HEATING UP TO 250°C
11	REFRIGERATOR (300 Litres)
12	THERMOMETERS (°C) Ranges: -10 to 50, -10 to 110, 0 to 150, 0 to 200 and 0 to 400)
13	WATER BATH
14	SP. GRAVITY BOTTLE, 10 ml & 25 ml
15	HEATING MANTLE
16	MICROSCOPE
17	SAMPLE CONTAINERS FOR NAPHTHA/ DIESEL
18	CHLOROFORM BELLOWS (Bladder Type)

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19	THERMOSTAT BATH
20	LPG CYLINDER WITH BURNER
21	WET & DRY BULB HYGROMETERS
22	VACUUM PUMP (DOUBLE STAGE)
23	MUFFLE FURNACE (UPTO 1200°C) FOR LOSS ON IGNITION

Note:

- i. All instruments mentioned in section 12.0 (other than those marked in #) are mandatory, site should keep stand-by to ensure uninterrupted testing facility.
- ii. Site should have separate oil particle counters for lubricating oil and control fluid analysis.
- iii. Dissolved Gas Analyser (for transformer oil)- stand-by instrument is mandatory for pit-head stations.
- iv. Additional instruments may be procured based on site-specific requirements.
- v. Site should maintain SOPs of instrument operation & calibration and records of calibration of all offline instruments with defined periodicity. *Illustrative template for instrument calibration is enclosed as Annexure-III.*
- vi. All critical instruments should be kept under annual maintenance contract for ensuring their 100% availability. Calibration frequency for instruments under AMC shall be as per OEM recommendations.
- vii. Instrument spares, reagents and sufficient stock of calibration standards is to be ensured and planned by site.

13.0 Online Instruments**A. Water Treatment Plant**

Table 13A				
S.N.	NAME OF THE EQUIPMENT	LOCATION OF SAMPLING POINT	ANNUNCIATION REQUIRED	REMARKS

TITLE	Code of Practices on Power Plant Chemistry
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1.	Conductivity Meter	a. Outlet of anion Exchanger ▪ Service ▪ Rinse b. Outlet of MB ▪ Service ▪ Rinse c. Make-up water line d. DM make-up pump discharge header	Yes Yes Yes Yes Yes Yes	On each DM stream --do-- On each DM stream --do-- MB Common header On each stream
2.	pH Meter	a. Outlet of MB b. Discharge line of effluent disposal pump of neutralization pit	Yes Yes	Low & high annunciation --do-- --do--
3.	Silica Analyzer	a. Outlet of anion Exchanger b. Outlet of MB	Yes	Common to all streams
4.	Sodium Analyzer	a. Outlet of SAC b. Outlet of MB	Yes	a. On each stream (DMP), if envisaged in design. b. For desal plant
5.	Conductivity Comparators	Outlet of SAC	Yes	On each stream (DMP), if envisaged in design.

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6.	Water Integrators & Flow meters	a. Inlet to clarifier b. Gravity filter outlet c. ACF outlet d. SAC outlet e. SBA outlet f. MB outlet g. Makeup & filling lines	-- -- -- -- -- -- --	
7.	Rotameters	a. Service inlet to all vessels b. Backwash line of all vessels c. Downflow & upflow lines of regenerant waterflow	-- -- --	
8.	Hydrometers & Online Concentration Analysers (for checking conc of regenerant)	a. After mixing tee for acid & alkali regenerants	--	If envisaged in design
9.	Level indicators (ultrasonic/ electronic type)	a. Water storage tanks b. Degasser sumps c. Acid/alkali storage and day tanks d. Alum/lime dosing tanks	-- -- -- --	
10.	DP Transmitters	a. ACF, WAC, SAC, WBA, SBA & MB vessels for all DM streams. b. Resin traps of ACF, WAC, SAC, WBA, SBA & MB vessels for all DM streams.		If envisaged in design

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B. Main Plant (Note: *Core parameters)
Table 13B

S.N.	Name Of The Equipment	Location of Sampling Point	Remarks
1.	FEED WATER ▪ pH METER ▪ CONDUCTIVITY METER (Sp.) ▪ CONDUCTIVITY METER (ACC)* ▪ DISSOLVED O2 ANALYZER* ▪ TOTAL IRON ANALYSER (Only for SC units, Talcher Kaniha & Simhadri)	Economizer Inlet	
2.	BOILER WATER (FOR DRUM TYPE UNITS) ▪ pH METER* ▪ CONDUCTIVITY METER (Sp.)* ▪ CONDUCTIVITY METER (ACC)* ▪ SILICA ANALYZER* ▪ CHLORIDE ANALYZER	Downcomer/ CBD Line/CC pump suction header	Phosphate analyzer not required in case of AVT/Caustic treatment.
3.	STEAM (SS/MS) ▪ pH METER ▪ CONDUCTIVITY METER (Sp.) ▪ CONDUCTIVITY METER (ACC)* ▪ SODIUM ANALYZER ▪ SILICA ANALYZER*	SS & MS Lines	Online Silica analyzer is required for LTSH Inlet & Main Steam samples in supercritical unit.

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4.	CONDENSATE ▪ pH METER ▪ CONDUCTIVITY METER (Sp.) ▪ CONDUCTIVITY METER (ACC)* ▪ DEGASSED CATION CONDUCTIVITY METER ▪ DISSOLVED O2 ANALYZER* ▪ SODIUM ANALYZER*	CEP Discharge	
5.	CPU OUTLET ▪ pH METER ▪ CONDUCTIVITY METER (Specific) ▪ CONDUCTIVITY METER (ACC) ▪ SODIUM ANALYZER ▪ DP AT EACH VESSEL OUTLET AND CPU INLET HEADER WITH TRANSMITTER ▪ DP AT RESIN TRAP OF EACH CPU VESSEL ▪ CPU FLOW TRANSMITTER WITH INTEGRATOR	CPU O/L	At each CPU service vessel. For details please refer OGN/OPS/CHEM /020 on "Guidelines on Operating Philosophy of Condensate Polishing Unit".
6.	DEAERATOR INLET • DISSOLVED O2 ANALYZER*	D/A Inlet	ORP is required for AVT(R) feed water treatments.
7.	MAKE UP WATER ▪ CONDUCTIVITY METER (Specific)* ▪ CONDUCTIVITY METER (ACC) ▪ DEGASSED CONDUCTIVITY METER (DCC)	Make-Up Line	

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8.	STATOR WATER CONDUCTIVITY METER (Sp.)	- SPU (Inlet & Outlet) - Winding inlet	For details refer OGN/OPS/CHEM /002 on "Chemical Control of Generator Stator Primary Water System".
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NOTES: Alarms shall be provided in the control room for core parameters, Annunciation is required for all parameters.

14.0 Equipment Inspection and Overhauling Activities

A. Equipment Inspection

Table 14A: Equipment Inspection Schedule & Work Description				
S.N.	EQUIPMENT	CHECK/WORK TO BE DONE	FREQUENCY	RESPONSIBILITY
1.	CLARIFIER	Structural Checks including concrete/civil structure. (Any Seepage issues, grating & railing availability) Scraper healthiness Launder healthiness in case of reactor clarifier Checking/overload test of of Rake drive, VFD drive based agitator, V belts. Healthiness of Desludging line& associated valves, telescopic valves. Painting in wetted portion/non wetted portion Coagulant and biocide dosing lines	Once/ year (before onset of monsoon)	MM-OFFSITE
2.	PUMPS & BLOWERS (in PT & DM Plant)	i. For all Vertical pumps/blowers	i. Six monthly inspection and OH in two years.	MM-OFFSITE
		ii. For all horizontal pumps/blowers.	ii. Three monthly inspection and OH in one year	

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3.	ALUM, LIME, COAGULANT- AID TANKS; AGITATORS	-do-	Once/ 6-months (cleaning)	MM-OFFSITE
			Once/ year (painting) or as required	
4.	GRAVITY/ PRESSURE/ ACTIVATED CARBON FILTERS/DMF/HIGH PRESSURE SAND FILTERS	Refer OGN-OPS-SYST- 018 on “Quality Management in Maintenance / Overhauling”.	Once/ 2-years or Earlier if required	For ACF and other closed vessel CHEMISTRY, MM-OFFSITE and FQA
5.	DEGASSER TOWERS/TANKS	Refer OGN-OPS-SYST- 018 on “Quality Management in Maintenance / Overhauling”.	As per OGN/OPS/SYST/01 8 on “Quality Management in Maintenance/ Overhauling”.	For tank lining MM-OFFSITE, CHEMISTRY and FQA; And for rest MM- OFFSITE, CHEMISTRY
5	DM PLANT VESSEL	Rubber lining inside the vessel- Visual Inspection and Hardness Testing (above the resin Level) Strainer Checking Supporting plate, Middle collector, lateral header inspection. Resin trap outlet header healthiness Caustic/Alkali dosing line and associated valves checking. Complete rubber- lining replacement is to be done if substantial portion of rubber hardness exceeds 85 Shore A/ Visual Cracks seen/ Major portion Sparks	Once/two years or as and when required- For WAC and SAC vessels Once/three years or as and when required, for all other vessels (except WAC and SAC)	MM-OFFSITE, CHEMISTRY and FQA;

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		observed during spark test.		
6	CPU Service & Regeneration Vessels	OGN-OPS-CHEM-020 on "Guidelines on Operating Philosophy of Condensate Polishing Unit).	As per OGN-OPS- CHEM-020 on "Guidelines on Operating Philosophy of Condensate Polishing Unit).	MM-OFFSITE, CHEMISTRY and FQA;
7	REGENERATION TANKS/ AGITATORS	Refer OGN-OPS-SYST-018 on "Quality Management in Maintenance / Overhauling"	Once/Year or earlier if required.	MM-OFFSITE, CHEMISTRY and FQA;
8	BULK ACID STORAGE TANKS	Inspection of tank nozzles and mounting of internals. Rubber lining healthiness checking as per OGN-OPS-CHEM-007. . Fume Absorber checking. . Manhole tightness and inspection. Valves associated with tanks	Annually or as and when required. For caustic soda tanks, frequency: once/two years	MM-OFFSITE, CHEMISTRY and FQA;
9	DM STORAGE TANKS	Refer OGN-OPS-SYST-018 on "Quality Management in Maintenance / Overhauling"	As per OGN-OPS- SYST-018 on "Quality Management in Maintenance / Overhauling".	MM-OFFSITE, CHEMISTRY and FQA;
10	CHLORINATION PLANT	Overhauling and replacement of corroded parts and lines. Proper painting and cleaning of system valves diaphragms	Overhaul once/ year. As and when required	MM-OFFSITE, CHEMISTRY

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		Protection/ interlock checks and control equipment checks. Any leakages	Six-monthly Immediate action	
11	CHLORINE CYLINDERS	Refer OGN/OPS/SYST/018 on "Quality Management in Maintenance / Overhauling"	As and when required (schedule to be fixed and maintained by site).	Chemistry
12	CHLORINE LEAK ABSORPTION SYSTEM (CLAS)	Refer OGN/OPS/SYST/018 on "Quality Management in Maintenance / Overhauling"	Once/month	MM-Offsite, Chemistry, Safety, C&I. For monthly mockdrills, CISF to be involved and joint protocols to be maintained

NOTE:

- i. Chemistry department shall coordinate with maintenance and FQA for vessel inspection as per schedule. Detailed Checklist of inspection and maintenance activities is to be recorded. Joint protocol is to be signed and maintained by all concerned departments involved in inspection.
- ii. Responsibilities (as mentioned against specific checks/inspection) may be modified based on site- specific requirements.
- iii. All concrete structures shall be inspected by O&M(Civil) at least once in every five years or earlier, as per site requirement.
- iv. All breathers and Silica gel to be checked every fortnight and replacement may be done as per requirement. Records for the same is to be maintained by site.

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Table 14B: Equipment Inspection Status**Name of the Station:** **Stage:****Year:**

Equipment type	Total nos available	Total nos inspected	Total nos inspection pending	Remarks
Clarifier				
Gravity Sand Filter/ High Pressure Sand Filter				
Filter Water Sump				
Drinking water/Potable water Sump				
Mixed Bed Vessel in DM Plant				
CPU Service Vessel				
CPU Regeneration vessel				
Acid Storage Tank				
Alkali Storage Tank				
DM Water Storage Tank				
Any Other (specify)				

B. Overhauling Jobs-Chemistry Section**Reference TC Documents:**

- i. OGN-OPS-SYST-018: Quality Management in Maintenance / Overhauling
- ii. OGN-OPS-SYST-015: Code of Practice on Power Plant Chemistry
- iii. OIN-OPS-CHEM-036: Guidelines for Sampling of Deposits from Steam Turbines & Components
- iv. OGN-OPS-CHEM-020: Guidelines on Operating Philosophy of Condensate Polishing Unit

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- v. OIN-OPS-SYST-087: Recommendations of Workshop on Online Instrumentation of SWAS & CPU System by Chemistry and C&I.

C. Overhauling Jobs-Chemistry Section

Reference TC Documents:

- i. OGN-OPS-SYST-018: Quality Management in Maintenance / Overhauling
- ii. OGN-OPS-SYST-015: Code of Practice on Power Plant Chemistry
- iii. OIN-OPS-CHEM-036: Guidelines for Sampling of Deposits from Steam Turbines & Components
- iv. OGN-OPS-CHEM-020: Guidelines on Operating Philosophy of Condensate Polishing Unit
- v. OIN-OPS-SYST-087: Recommendations of Workshop on Online Instrumentation of SWAS & CPU System by Chemistry and C&I.

Table 14C: Chemistry Overhauling Activities Check-List			
Name of the station:			
Unit: Period of overhauling:			
Sl.No	Description of Checks, Inspections & Activities	Responsibility	Remarks
A	CPU		
1	Inspection of all CPU service vessels & Resin trap (rubber lining inspection including hardness test is to be ensured)	TMD, Opn & Chem, FQA	Refer OGN-OPS-CHEM-020
2	Calibration of pressure transmitter, switches and Pressure Gauges of CPU service vessel area.	C&I	
3	Calibration of flow meters of all CPU service vessels.	C&I	
5	CPU air blowers servicing.	TMD, EMD	
6	Inspection of all manual & pneumatic valves.	TMD	
7	Position Calibration of auto bypass valve.	C&I, Chem, Opn	
8	Protection interlock checking of CPU service vessels and bypass valves as per latest COS recommendations.	C&I, Chem, Opn	

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9	DMW storage tank inspection.	MM-Offsite, Chem, FQA	
10	Any other shut-down job	Chem	
B	SWAS		
i	Primary Rack		
1	Cleaning/painting of Rack	BMD	
2	Cleaning/servicing of all isolation valves/PRVs	BMD/TMD	
3	Cleaning/servicing of all local instruments	C&I	
4	Cleaning/servicing of primary coolers	BMD	
5	Healthiness checks and replacement of Sample pipelines for sample upto primary cooler	BMD	
ii	Secondary Panels		
1	Cleaning/painting of wet panel	C&I	
2	Cleaning/painting of dry panel	C&I	
3	Cleaning/servicing of all rotameters	C&I	
4	Cleaning/servicing of secondary coolers	C&I	
5	Cleaning of pH sensors & chambers	C&I	
6	Calibration of PH analyzers	C&I	
7	Cleaning of Specific Conductivity sensors & chambers	C&I	
8	Calibration of Specific Conductivity analyzers	C&I/Chem	
9	Cleaning of Cation Conductivity sensors & chambers	C&I	
10	Calibration of Cation Conductivity analyzers	C&I/Chem	
11	Cleaning of Dissolved oxygen sensors & chambers	C&I	
12	Calibration of Dissolved oxygen analyzers	C&I/Chem	
13	Cleaning of ORP sensor & chamber	C&I	
14	Calibration of ORP analyzer	C&I/Chem	

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15	Cleaning of Silica, Phosphate, Sodium, Chloride sensors & Chambers	C&I	
16	Calibration of Silica, Phosphate, Sodium, hydrazine & Chloride analyzer	C&I/Chem	
17	Calibration of temperature switches	C&I	
18	Calibration of temperature gauges	C&I	
19	Calibration of pressure gauges	C&I	
20	SWAS lighting	EMD/C&I	
21	Proper cable dressing	C&I	
iii	SWAS Chiller System		
1	SWAS Cooler (Primary & secondary) checking/ Cleaning with valves & filters.	C&I	
2	Sample chiller compressor overhauling & Compressor gas filling.	TMD/ MMD-OFS	
3	Calibration of pressure gauge, temp gauge & switches.	C&I	
4	Flow switch servicing, calibration & checking	C&I	
5	Temperature controller healthiness checking	C&I	
6	Local panel healthiness checking	C&I	
7	Cooler cleaning of local sampling rack.	MMD	
8	Maintenance of sample root valves.	MMD	
iv	Any other shut-down job	Chem to coordinate with mtn group.	
C	Boiler		
1	Boiler water wall tube collection- High heat zone (12nos) for analysis from CC-NETRA (Pre-cleaning and post cleaning inspection and photos). Preliminary tests of deposit to be done at site.	BMD & Chem	
2	Boiler separator sample collection (Pre-cleaning and post cleaning inspection & photos). Preliminary tests of deposit to be done at site.	BMD & Chem	

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3	CBD valves checking for passing, if any. CBD root valves servicing, CBD sample line thorough cleaning and deposit removal, if any.	BMD	
4	Any other shut-down job	Chem to coordinate	

D	Condenser, Hot well and CST		
1	- Inspection of Condenser tube, water box- Pre & post cleaning inspection and photos. - Boroscopic inspection of condenser tubes.	TMD, EEMG & Chem	
2	Preliminary tests of deposit to be done at site. Further sample to be sent to NETRA.		
3	Inspection and servicing of condensate dumping valve	TMD	
4	Condenser flood test: - To be witnessed by site chemistry - Records of number of tubes plugged in current OH (pass- wise) - Records of number of tubes plugged in previous OH (pass- wise) to be maintained	TMD, Opn, EEMG & Chem	
5	Painting of water box (if required)	TMD	
6	Condenser air-in leakage to be identified and acted upon during overhaul.	TMD	
7	Pre-cleaning and post cleaning inspection of Deaerator, Hot well, MOT.	TMD, Opn & Chem	
8	Hot well conductivity meter cleaning & calibration.	C&I	
9	Make-up spray & air evacuation system in hot well	TMD, Opn	

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	with Vacuum pump.		
10	CST inspection.	TMD, Opn & Chem	
11	Any other shut-down job	Chem to coordinate with mtnc group.	
E	Turbine		
1	- During capital overhauling, turbine inspection and deposit sample collection to be done by chemistry along with TMD. - The stages where deposition occurs also to be recorded. - Preliminary tests of deposit are to be done at site. - Further samples are to be sent to CC-NETRA.	TMD, Chem	
2	Any other shut-down job	Chem	
F	Generator Stator Primary water system		
1	Calibration of online conductivity & DO analysers	C&I	
2	Cleaning of Alkali dosing tank.	TMD	
3	SPU resin changeover	Chem, Opn, EMD	
4	Any other shut-down job		
G	Boiler chemical dosing system		
1	Cleaning of LP-HP chemical dosing storage/day tanks.	MMD	
2	LP, HP pumps overhauling.	MMD	
3	Pressure gauge calibration.	C&I	
4	Calibration of stroke controller of all dosing pumps.	TMD, C&I	
5	All valves checking, including the one near boiler drum.	TMD	

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6	Checking of local control panel.	C&I	
7	Any other shut-down job	Chem to coordinate with mtnc group.	
J	Other Miscellaneous jobs		
1	UPS supply redundancy checking in systems	C&I	
2	Healthiness checking of all parameters in DCS and PI system before start of unit recommissioning activities.	C&I	

Note:

- i. Copies of Joint Protocol for all inspection/checks/activities are to be maintained and signed by concerned groups; copies to be uploaded on site intranet/mailed to COS-Chemistry for reference.
- ii. Responsibilities (as mentioned against specific jobs) may be modified based on site-specific requirements.

15.0 RECORDS TO BE MAINTAINED

S.N.	Record Description	Maintained by	Frequency	Remarks
1	As per tables in sections 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0, 12.0, 13.0, 14.0, Annexures-I&II.	Chemistry	As mentioned in the relevant sections	

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16. RESPONSIBILITY

S. No.	Code of Practice	Responsibility
4.2	Summary of COS Guidelines issued	DGM(Chem)
5	Water Treatment Plant: Parameter Testing and Operational Records	
5.1	Raw Water/ Clarified water/ Filtered water/ Drinking water	Water Lab In-charge
5.2	Guidelines for operation and maintenance of resin-based DM Plant	Shift Chemist
5.3	Water production/ consumption data	Shift Chemist
5.4	DM Water and clarified water cost calculations	Shift Chemist
5.5	DM plant performance data (monthly updation)	Shift Chemist
5.6	Bulk chemical consumption	Water Lab In-charge
5.7	Specific chemical consumption data for DM Plant	Water Lab In-charge
5.8	G. Activated Carbon/ Resin Testing (DM Plant, CPU)	Water Lab In-charge
5.9	Sample log sheets	Shift Chemist
6	Main plant: Parameters to be tested/ recorded for steam water cycle and their normal frequency of testing/ recording	Shift Chemistry
7	Cooling water system	ENV. Lab In-charge
8	Coal and Fuel Analysis	Coal Lab In-charge
	J. Coal-As-Received- Indigenous Coal	
	K. Coal-As-Received- Imported Coal	
	L. Coal as Fired	
	M. Ultimate analysis & Ash analysis	
	N. Mill Fineness	
	O. Mill reject Analysis	
	P. Combustibles in bottom ash and fly ash (dry basis)	
	Q. Coal stock-pile GCV mapping	
	R. Biomass pellets analysis	
9	Limestone and Gypsum Analysis For FGD	General Shift Executive
10	Oil Analysis	Oil Lab In-charge
	D. Fuel oil Analysis	
	E. Lube oil and control fluid analysis	
	F. Transformer oil Dissolved gas analysis	
11	Gas Analysis	Shift Chemist
	D. Flue Gas Analysis	
	E. Hydrogen gas analysis	
	F. Instrument Air	Water Lab In-charge
12	Laboratory Equipment	

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	D. Water Analysis Instruments	Water Lab In-charge
	E. Coal Analysis Instruments	Coal Lab In-charge
	F. Oil Analysis Instruments	Oil Lab In-charge
	D. Instruments for flue gas, DGA and hydrogen purity	Oil Lab In-charge
	E. Cooling water analysis instruments	ENV Lab In-charge
	F. Environmental monitoring instruments	Water Lab In-charge
	G. General laboratory instruments/ equipment	Water Lab In-charge
13	Online Instruments	
	C. Water Treatment Plant	Shift Chemist
	D. Main Plant (Note: * core parameters)	SWAS lab In-Charge /Shift chemist
14	Equipment inspection and Overhauling activities	Maint. Coordination Executive
	C. Equipment inspection	
	D. Overhauling Jobs- Chemistry Section	
15	Records	DGM(Chem)

17. REFERENCE DOCUMENT:

COS-ISO-00-OGN/OPS/CHEM/015 Rev. No.: 04 Dec. 2025

18. REVIEW:

This document will be reviewed by Head of Chemistry once in two years.

19. Checklist

Annexure – I: Equipment non-availability and troubleshooting

Annexure – II: Clarified Water and DM Water Cost Calculation Template (Illustrative)

Annexure – III: Illustrative template for Chemistry laboratory instrument calibration

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Annexure – I: Equipment non-availability and troubleshooting

(FREQUENCY: Monthly updation to be done)

STATION/PROJECT:

S.N.	Date	Area	System / Equipment not available for >15 days	Action taken	Responsibility	Target Date	Current Status

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Annexure – II: Clarified Water and DM Water Cost Calculation Template (Illustrative)

Name of the station:	
Month-Year:	

A. Pre-Treatment Cost:

Raw water Charges:

Sr No	Particulars	UOM	Quantity (M3)	Water Charges/M3 (INR)	Total Cost (INR)
1	Total raw water treated in PT Plant (DM Clarifier)	M3			
Pre-Treatment Chemical cost:					
Sr No	Chemical consumption	UOM	Quantity	Landed Cost of chemicals (INR)	Total Cost (INR)
1	Ferric Alum Gr4- Coagulant	MT			
2	PAC- Coagulant	MT			
3	Hydrated Lime	MT			
4	Chlorine- Biocide	MT			
5	Sodium hypochlorite- Biocide	MT			
6	Polyelectrolyte/Polydadmec	MT			
9	Hydrochloric acid (30%) for ClO ₂ generation	MT			
Total PT Chemical Cost (INR)					
Pre-Treatment Chemical Unit Cost (Rs / M3)					
PTP Man-Power Cost (INR)					

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Pre-Treatment Man-Power Unit Cost (Rs / M3)	
PTP Power Cost (INR)	
Pre-Treatment Power Consumption Unit Cost (Rs / M3)	
Pre-Treatment Unit Cost (Rs / M3)	

B. DM Water Production Cost:

Total DM Water production (M3)	
---------------------------------------	--

DMW production chemical cost:

Sr No	Chemical consumption	UOM	Quantity	Landed Cost of chemicals (INR)	Total Cost (INR)
1	Hydrochloric acid (HCl)- 30%-Regenerant	MT			
2	Caustic Soda (NaOH)- Lye - 100%-Regenerant	MT			
3	Caustic Soda (NaOH)- Flakes- 100%-Regenerant	MT			
4	Any other (site-specific)				

Total Chemical Cost for DMW Production (INR)	
DMW Production chemical cost/M3 (excluding PTP cost)	
DMP Man-Power Cost (INR) excluding PTP cost	
DMW Production Man-Power cost/M3 (excluding PTP cost)	
\$DMP Power Cost (INR) excluding PTP cost	
DMW Production Power cons cost/M3 (excluding PTP cost)	
*Cost of MB resin (INR)	

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DMW Production MB Resin cost/M3 (excluding PTP cost)	
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DMW Unit Cost (Rs/M3) excluding pre-treatment cost & RW cost	
---	--

SUMMARY:

Raw water Cost (Rs/M3)	
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Pre-Treatment Unit Cost (Rs/M3) including man-power cost, chemical cost, power cons cost	
---	--

Pre-Treatment Unit Cost (Rs/M3) excluding man-power cost	
---	--

Total Clarified water production Unit Cost (Rs/M3) including man-power cost	
--	--

Total Clarified water production Unit Cost (Rs/M3) excluding man-power cost	
--	--

Total DM water production Unit Cost (Rs /M3) including raw water cess, man-power cost, power cons cost, chemical cost, resin cost, cost of membranes, cost of cartridge	
--	--

Total DM water production Unit Cost (Rs /M3) excluding man-power cost	
--	--

Total DM water production Unit Cost (Rs /M3) excluding man-power cost and raw water cess	
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Note:

\$ Power cost for pumps (power consumption x running hours).

*Resin- 10% of total cost per year to be considered.

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Annexure – III: Illustrative template for Chemistry laboratory instrument calibration

Name of the Station:

Month:

S.I.N o.	Type of Instrum ent	Mak e/ Mod el/ SI No.	Unique Identifi ca tion No. (UIN)	SOP refere nce no. for calibr at ion	Calibr at ion Frequ e ncy	Date of Calibr at ion	Stan d ard used	Devia t ion obser v ed (If any)	Accep ta nce Criteri a	Rem a rks	Calibrate d By		Reviewed By	
											Na me	Signa ture	Na me	Signa ture

Note:

- Separate records are to be maintained for respective instrument types.
- Record of list of calibration standards along with validity dates is to be maintained.